

Instalasi Webserver Menggunakan Virtual Machine

Panduan Konfigurasi Jaringan Antar 2 VM

 VirtualBox

 Apache2

 Linux

| Pengenalan Virtualisasi & Web Server



Virtual Machine (VM)

- ✓ Komputer virtual yang berjalan di atas sistem fisik
- ✓ Memungkinkan simulasi jaringan tanpa banyak perangkat keras
- ✓ Isolasi sistem yang aman untuk uji coba



Web Server Apache

- ✓ Perangkat lunak yang melayani permintaan HTTP
- ✓ Mengirimkan halaman web ke browser client
- ✓ Open-source dan stabil untuk produksi

🚩 Tujuan Praktikum

Membangun simulasi arsitektur **Client-Server** menggunakan dua buah Virtual Machine yang terhubung dalam satu jaringan virtual untuk memahami cara kerja web server secara langsung.

| Persiapan Lingkungan

Software yang Dibutuhkan



Oracle VirtualBox

Software virtualisasi gratis



ISO Linux Server

Ubuntu Server 20.04+ / Debian



SSH Client (Opsional)

Putty / Terminal Linux

Spesifikasi Minimum VM



RAM

512 MB+



Storage (HDD)

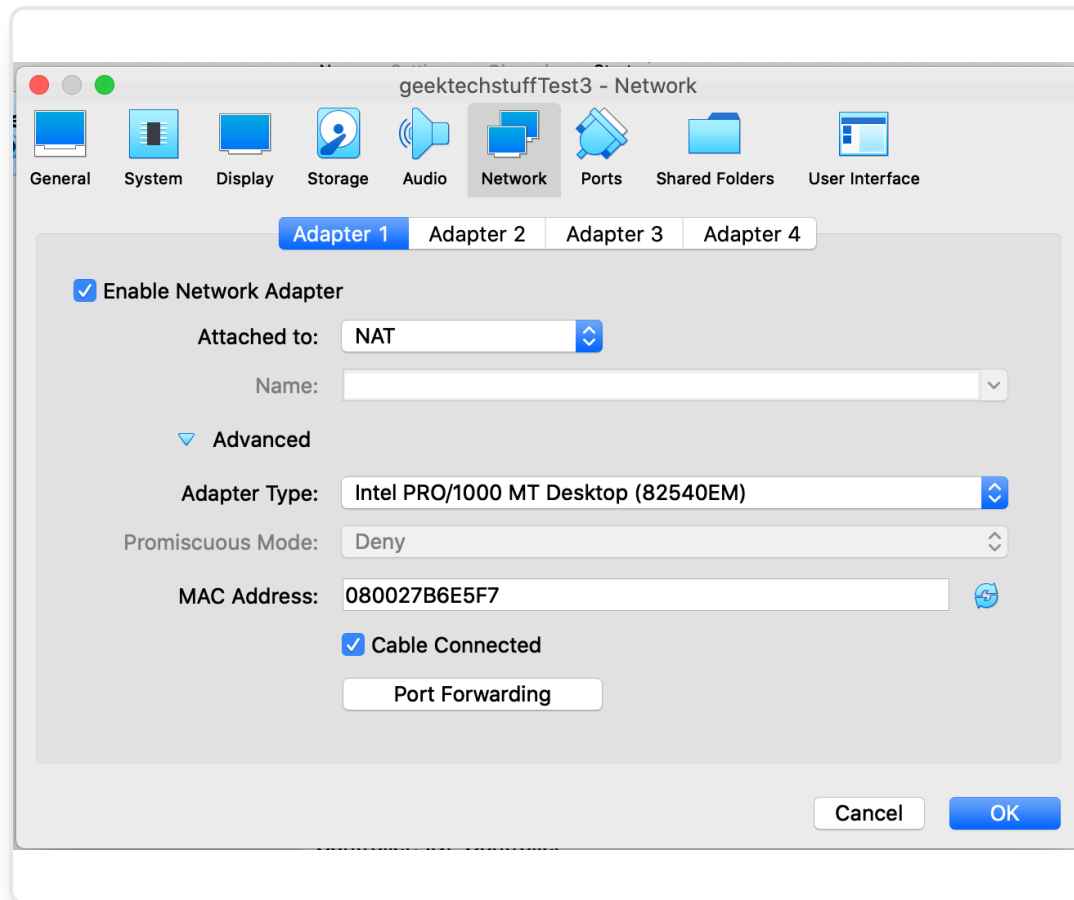
10 GB+



Network Adapter

Internal Network

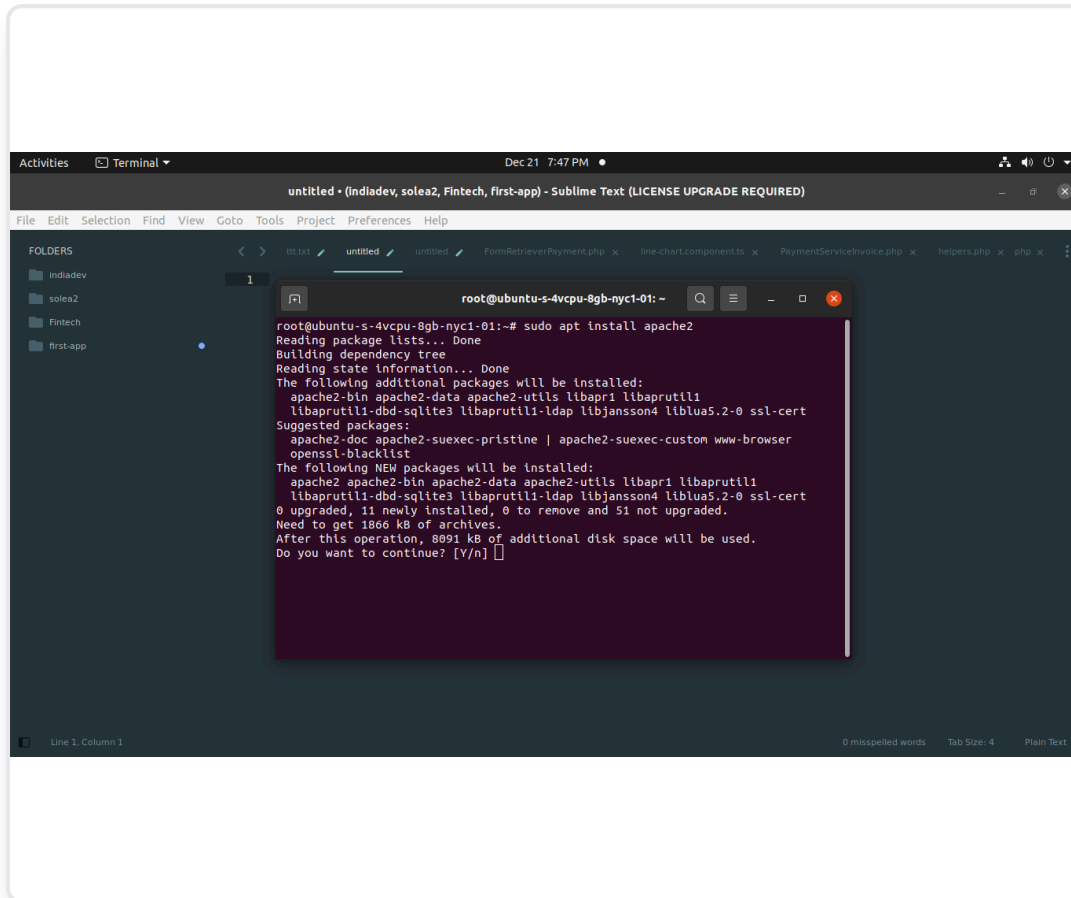
Konfigurasi Jaringan VirtualBox



Mode Internal Network

- 1 Buka Settings VM**
Klik kanan VM → Settings
- 2 Pilih Network**
Tab Network → Adapter 1
- 3 Atur ke Internal Network**
Attached to: **Internal Network**
- 4 Nama Jaringan Sama**
Pastikan kedua VM menggunakan nama jaringan yang identik

Instalasi Web Server Apache



```
root@ubuntu-s-4vcpu-8gb-nyc1-01:~# sudo apt install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
  openssl-blacklist
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0 ssl-cert
0 upgraded, 11 newly installed, 0 to remove and 51 not upgraded.
Need to get 1866 kB of archives.
After this operation, 8091 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Langkah Instalasi

1 Update Repository

```
sudo apt update
```

2 Install Apache2

```
sudo apt install apache2 -y
```

3 Verifikasi Status

```
sudo systemctl status apache2
```

i Default Port: HTTP port 80

Konfigurasi IP Statis - Server

↔ VM Server - Konfigurasi Jaringan

 IP Address

192.168.1.10/24

 File Konfigurasi

Path:

/etc/netplan/01-network-manager-all.yaml

► Terapkan Konfigurasi

sudo netplan apply

```
ostechnix@ubuntu2204:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0e:26:0b:0e:33:8c brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 192.168.1.40/24 brd 192.168.1.255 scope global noprefixroute ens18
        valid_lft forever preferred_lft forever
    inet6 fe80::bfcf:b9d3:60de:af1b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ostechnix@ubuntu2204:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0e:26:0b:0e:33:8c brd ff:ff:ff:ff:ff:ff
    altname enp0s18
ostechnix@ubuntu2204:~$
```

| Konfigurasi IP Statis - Client

VM Client - Konfigurasi Jaringan

 IP Address

192.168.1.20/24

 **Penting!**

Subnet Mask: 255.255.255.0

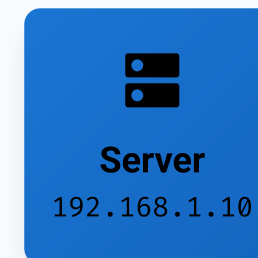
Gateway: Opsional (192.168.1.1)

 **Verifikasi Jaringan**

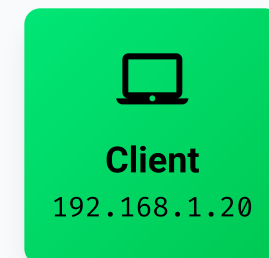
Pastikan Server (192.168.1.10) dan Client (192.168.1.20) berada dalam **subnet yang sama**

Topologi Jaringan Internal

 Internal Network




Ping Test



| Pengujian Koneksi Antar VM

Uji Konektivitas

1 Akses Terminal

Buka terminal pada **VM Client**

2 Jalankan Perintah Ping

```
ping 192.168.1.10
```

✓ Hasil Berhasil

Jika muncul "Reply from 192.168.1.10", koneksi berhasil!

Terminal - VM Client

```
$ ping 192.168.1.10

PING 192.168.1.10 (192.168.1.10) 56(84)
bytes of data.

64 bytes from 192.168.1.10: icmp_seq=1
ttl=64 time=0.042 ms

64 bytes from 192.168.1.10: icmp_seq=2
ttl=64 time=0.085 ms

64 bytes from 192.168.1.10: icmp_seq=3
ttl=64 time=0.093 ms

64 bytes from 192.168.1.10: icmp_seq=4
ttl=64 time=0.071 ms

$ █
```

✓ **Koneksi Berhasil!**

Akses Web Server dari Client

🌐 Langkah Akses

1 Buka Browser

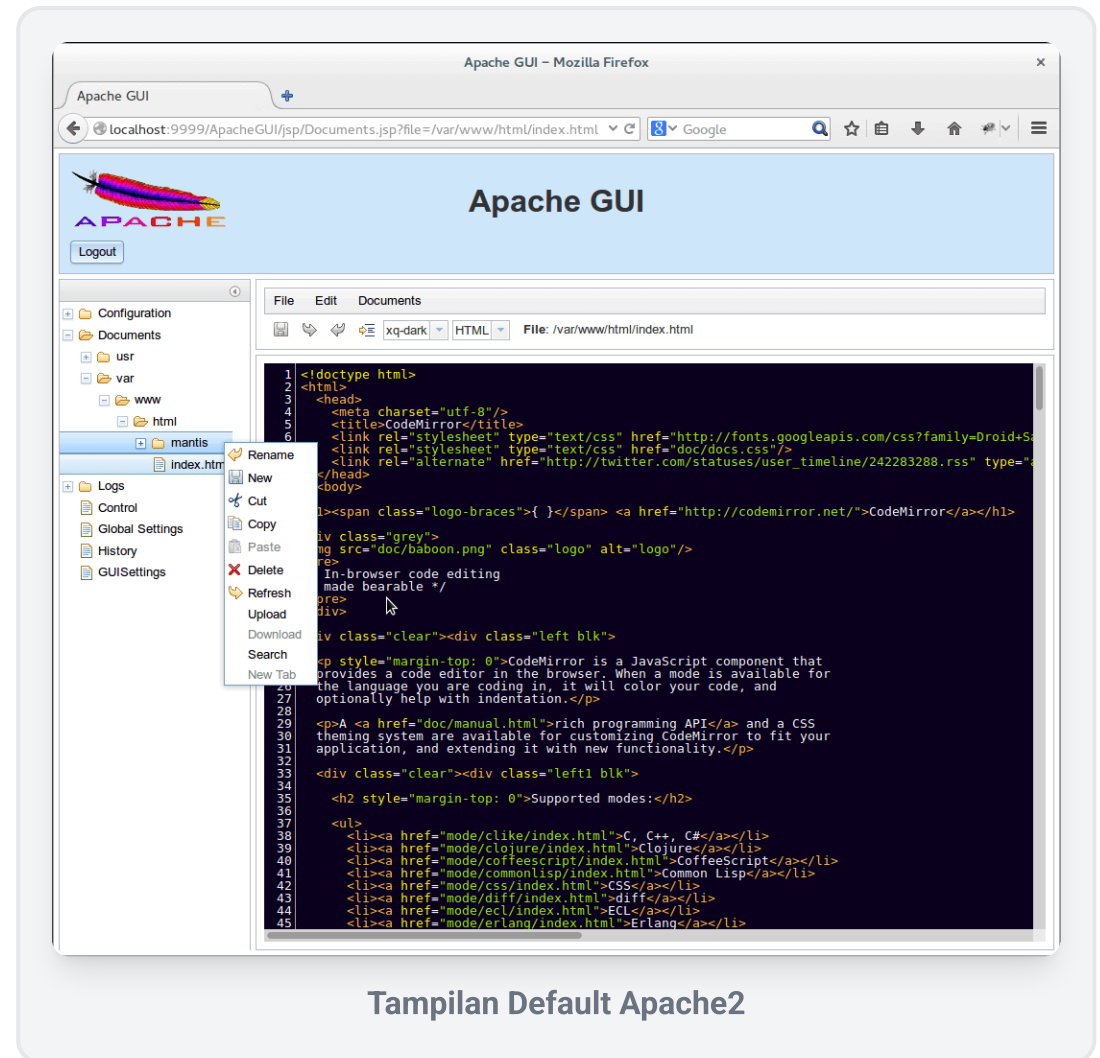
Jalankan browser di VM Client

2 Masukkan Alamat IP

HTTP `http://192.168.1.10`

✅ Berhasil!

Halaman default Apache2 "It works!" akan muncul



Modifikasi Halaman Web

Langkah Edit

1 Akses Terminal

Buka terminal di VM Server

2 Buka File Index

```
nano /var/www/html/index.html
```

3 Edit Konten

Ubah teks sesuai keinginan, lalu simpan dengan
Ctrl+X, Y, Enter

Refresh Browser

Lihat perubahan di VM Client

nano - /var/www/html/index.html

```
1 <html>
2 <head><title>Web Server Saya</title></head>
3 <body>
4 <h1>Selamat Datang!</h1>
5 <p>Web server berhasil dikonfigurasi.</p>
6 <ul>
7 <li>Ubuntu Server 20.04</li>
8 <li>Apache2 Web Server</li>
9 </ul>
10 </body>
11 </html>
```

✓ File Tersimpan

UTF-8 | LF | HTML